

Updating the Data

1. One-Line Configuration using `addAnnotatedClass()`:

- Replaces the need for `hibernate.cfg.xml`.

```
SessionFactory sessionFactory = new  
Configuration().configure().addAnnotatedClass(Student.class).buildSessionFactory();
```

Overall Workflow:

1. `new Configuration()`: Creates a configuration object.
2. `configure()`: Loads the Hibernate configuration details.
3. `addAnnotatedClass(Student.class)`: Registers the `Student` class as an entity.
4. `buildSessionFactory()`: Builds the `SessionFactory`, which is used to create sessions for interacting with the database.

2. `update()`:

- Used to update an existing record in the database.
- The entity must already exist in the database (identified by its ID).
- Throws an exception if the entity is not already in the database.
- Example:

```
session.update(student);
```

3. `saveOrUpdate()`:

- Saves a new entity or updates an existing one based on its identifier (ID).
- If the entity is new (no ID in the database), it inserts. If it exists, it updates.
- Example:

```
session.saveOrUpdate(student);
```

4. **merge()**:

- Similar to `saveOrUpdate()`, but works with detached entities (not in session).
- Returns a new instance that is managed by Hibernate; does not directly update the original object.
- Useful when working with multiple sessions.
- Example:

```
Student updatedStudent = session.merge(student);
```

Key Differences:

- **save()**: Only inserts new records. Throws an exception if the record exists.
- **saveOrUpdate()**: Inserts if new, updates if the record already exists.
- **merge()**: Updates detached entities and works with a new instance, does not modify the original entity directly.

Code:

```
import org.hibernate.cfg.Configuration;
import jakarta.persistence.EntityExistsException;
import org.hibernate.HibernateException;
import org.hibernate.Session;
import org.hibernate.SessionFactory;
import org.hibernate.Transaction;
import org.hibernate.exception.ConstraintViolationException;
public class Main {
    public static void main(String[] args) {
        Configuration config = null;
        SessionFactory sessionFactory = null;
        Session session = null;
        Transaction transaction = null;
        boolean flag = false;
        try {
            config = new Configuration();
```

```

    config.configure();
    sessionFactory = config.buildSessionFactory();
    session = sessionFactory.openSession();
    // Detached entity that we want to update in the database
    Student student = new Student();
    student.setId(134588); // Make sure this ID exists in the database for an
update
    student.setName("Shiva Updated with Merge");
    student.setCity("New City Ballia");
    transaction = session.beginTransaction();
//    session.update(student);
//    session.saveOrUpdate(student);
    Student updatedStudent = session.merge(student); // Merge operation
    transaction.commit();
    System.out.println("Transaction committed successfully using merge.");
    flag = true;
} catch (EntityExistsException e) {
    System.out.println("Entity already exists: " + e.getMessage());
} catch (ConstraintViolationException e) {
    System.out.println("Constraint violation: " + e.getMessage());
} catch (HibernateException e) {
    System.out.println("Hibernate exception: " + e.getMessage());
} catch (Exception e) {
    System.out.println("Unexpected exception: " + e.getMessage());
} finally {
    if (!flag) {
        if (transaction != null) {
            transaction.rollback();
            System.out.println("Transaction rolled back successfully");
        }
    }
    if (session != null) {
        session.close();
    }
}

```

```

        if (sessionFactory != null) {
            sessionFactory.close();
        }
    }
}
}

```

Result:

Zero param constructor

Hibernate:

```

select
    s1_0.sid,
    s1_0.scity,
    s1_0.sname
from
    StudentTable s1_0
where
    s1_0.sid=?

```

Zero param constructor

Transaction committed successfully using merge.

```

mysql> select * from studenttable;
+-----+-----+-----+
| sid   | scity          | sname                      |
+-----+-----+-----+
| 13    | Ballia         | Arjun                      |
| 123   | Ballia         | Shiva                      |
| 134   | Ballia         | Arjun                      |
| 1233  | Ballia         | Arjun                      |
| 1345  | Ballia         | Shiva                      |
| 13457 | Ballia         | Shiva                      |
| 134588 | New City Ballia | Shiva Updated with Merge |
+-----+-----+-----+
7 rows in set (0.00 sec)

```